Presentation 2

What did each of the team members do?

Note that multiple team members can work together on the same task. In that case, you may write the same task for all of them. For example,

Aidan: Made mandatory code changes and ran code to get results after mandatory changes. Worked on task of splitting benign cases with too many slices into more, new benign cases. Ran code after successfully splitting 5 of the 7 original benign cases into a total of 11 new cases.

Paris: Spent time understanding how Optuna could actually be implemented into the code base, and then wrote a python script to implement changes. Spent time debugging this and working on a strategy to use Optuna but still have it run in a reasonable amount of time. Ran the model with the Optuna-determined hyperparameter settings and analyzed results, proposed change for future implementation.

Ronaldo: Analyzed effective patch attention graphs and slice-level by stain visualizations to understand how the model was performing with the new split patches, specifically in regards to benign cases.

Fauzan: Implemented entropy regularization to the attention MLPs, ran the model but I made a mistake when running, so results were suboptimal. Drafted and mathematically outlined how we should use dropout throughout the model, proposing to reduce/remove the dropout in our attention MLPs, especially if we're implementing the entropy regularization.

Emma: Analyzed the top and bottom effectiveness and attention levels of the patches from the visualizations to understand what could be causing incorrect classifications.

How does it help the project?

Aidan: Splitting the benign cases with >5 slices into new cases will allow the model to use more benign case data than it was previously. The goal of this is to help the model more accurately predict the class of the benign cases by seeing more benign case data.

Paris: Though this implementation didn't improve performance, it ideally will in the future once we nail down the best usage of it and what hyperparameters to focus on. Ideally will help to make a more accurate model if run over more trials with more parameters.

Emma: Get some insight into what patterns the model could be learning and incorrect patterns it is learning in order to correct for them. Also learned that there are small blurry patches without any helpful details in them that the model is attending to.

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Ronaldo: As the split patch approach didn't make any negative impact to our model (as shown through the visualizations I analyzed), we can move forward with this approach and combine it with other methods (like Optuna) to hopefully improve model performance

Fauzan: The entropy implementation + the dropout recommendation should make our model more robust to peaky attention distributions that are bad at predicting benign cases.

Issues faced that haven't been resolved:

Aidan: Still need to split case 82 into two cases by splitting sox10 patches. Found that case 27 is missing H&E and Sox10 patches in its match 2, despite these slices existing in OneDrive.

Paris: Optuna did not make a major impact, which I theorized is related to the class splits. Predicted everything as high-grade, so need to shift parameter focus to avoid this.

Emma: How to account for empty spaces within the patches to avoid the model identifying and paying attention to those rather than the actual cells.

Fauzan: Need to run the model with entropy properly, and see its effect on attention distributions, with a focus on performance on benign cases.

Proposed next steps:

Paris: Run Optuna again including class weights and see if this improves performance, integrate with all other code changes made throughout the week.

Fauzan: Train the entropy-enabled model again after correcting and do analysis on the results. Look at model performance with/without dropout in the attention MLPs to see if their removal/reduction have a positive impact on performance.

References